Characterizing the Detectability Landscape of Iterative Model Collapse

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Abstract

When language models train iteratively on their own output, the resulting distributional collapse is difficult to detect: the time series are short, the signal grows sub-linearly with contamination, and—as we demonstrate—the choice of statistical method can shift conclusions from universal detection to complete null. We apply five lead-lag detection methods to identical iterative-training data and find that null-control false-positive rates span 7% to 94%, exposing method sensitivity as a primary obstacle to reliable collapse monitoring. Permutation Granger causality, the lowest false-positive method, reveals a structured detectability landscape: detection jumps from absent to reliable within a narrow contamination interval, organized by a power-law boundary across three confirmed regimes. Cross-architecture evaluation across five model families supports forward detection in the majority of conditions. Token entropy Granger-precedes diversity collapse by at least one generation, providing a canary signal whose triggered cessation reduces peak perplexity in a proof-of-concept demonstration.

1 Introduction

When language models train iteratively on their own output, the training distribution progressively loses its tail modes, a phenomenon termed model collapse (Shumailov et al., 2024; Alemohammad et al., 2024). This process can be triggered by vanishingly small synthetic fractions (Dohmatob et al., 2025): lexical diversity narrows, calibration deteriorates, and each generation inherits the biases of its predecessors. As machine-generated text grows prevalent on the web, such feedback loops pose a growing risk to production pipelines, and by the time standard quality metrics register degradation, the distribution may have already shifted irreversibly.

This threat motivates a fundamental question: *can we reliably detect the onset of model collapse?* The answer is less clear than it might appear. Iterative training produces short, noisy time series (often fewer than 15 points) where detection power drops sharply (Boettiger and Hastings, 2012), and model collapse is a first-order transition (Truong and Truong, 2026), structurally ruling out the critical-slowing-down early-warning signals that ecologists rely on (Scheffer et al., 2009) (§2). More fundamentally, no prior work has systematically compared detection methods on this problem; as we show, the choice of statistical test alone can shift conclusions from universal detection to complete null on identical data.

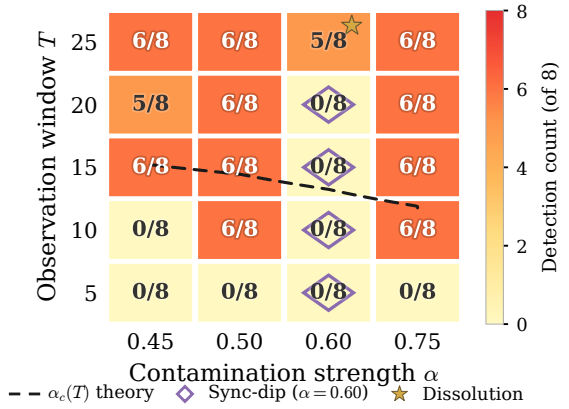


Figure 1: Phase diagram (Pythia-410M, fp32, PermG). FDR-significant canary→diversity pairs across α and T; dashed curve: fitted $\alpha_c(T) \propto T^{-\gamma}$; ⋄: non-monotonic anomaly; ⋆: dissolution at T=25.

We systematically map this landscape on Pythia-410M (Biderman et al., 2023) across 13 contamination levels with multi-seed replication (Figure 1). Our most consequential finding concerns method sensitivity: five statistical methods applied to identical data yield null-control FPR spanning 7% to 94% (κ=0.185; Figure 2), demonstrating that single-method analyses cannot reliably distinguish real signals from artifacts. Permutation Granger causality (PermG), the lowest-FPR method (7.4%, CI

$[2.1\%, 17.9\%]$, 9 seeds), reveals structured terrain rather than smooth degradation: detection is absent through α=0.45, then jumps from 0/8 to 6/8 significant pairs within a 2-percentage-point interval (Figure 1), before a non-monotonic dip and eventual reliable detection above a power-law boundary $\alpha_c(T)$ (Eq. 2). Cross-architecture evaluation at α=1.0 across five families (345M–6.9B) supports forward detection in 11/15 conditions, with four non-confirming cases isolating distinct failure modes (§4.5). Our canary claim is scoped to diversity metrics (distinct-n (Li et al., 2016)); Granger precedence establishes temporal predictive improvement, not causation (§3).

Our contributions are:

1. **We document method sensitivity as a primary finding** (§3.3). Five methods yield null-control FPR from 7% to 94% (κ=0.185); PermG, the lowest-FPR method across all five (7.4%, CI $[2.1\%, 17.9\%]$, 9 seeds), is selected as the primary detector.

2. **We map the detectability landscape** of iterative model collapse (§4.2–4.3). Detection is absent through α=0.45, then jumps from 0/8 to 6/8 within 2 percentage points; a detection boundary consistent with power-law scaling $\alpha_c(T) \propto T^{-\gamma}$ ($\gamma \in [0.64, 2.08]$; Eq. 2) organizes three confirmed regimes (blind, onset, detection) plus one unexplained non-monotonic anomaly.

3. **We empirically validate a cascade ordering for canary metric selection** (§3.1). Under mode amplification, entropy responds before diversity by $\geq$1 generation (Principles 1–2), confirmed in 14/15 experiments ($\alpha \geq 0.30$; GPT-2-medium s43 bidirectional).

2 Related Work

2.1 Model Collapse: Theory and Empirical Evidence

Iterative training collapse was formalized by Shumailov et al. (2024) and Alemohammad et al. (2024); Schaeffer et al. (2025) systematize eight distinct notions of collapse—our operationalization aligns with their *distributional shift* family (definitions D4–D6). Theoretical analyses sharpen these results: Dohmatob et al. (2024) recast collapse as changed scaling-law exponents, Dohmatob et al. (2025) proved collapse at vanishingly small synthetic fractions, and Truong and Truong (2026) prove entropy collapse is a first-order transition, structurally ruling out critical-slowing-down early-warning signals (Scheffer et al., 2009, 2012). Critical mixture ratios separating qualitative outcomes have independent precedent: Kazdan et al. (2025) map collapse conditions, Gu et al. (2024) establish scaling laws for critical ratios, and Gu et al. (2025) show data mixing induces phase transitions in knowledge acquisition.

2.2 Detection of Synthetic Data Contamination

Monitoring metrics proposed for collapse include expected calibration error (ECE) (Du et al., 2025), surplexity (Gambetta et al., 2024), and machine-generated text detection (Drayson et al., 2025); Shi et al. (2025) independently validate entropy as a collapse diagnostic in diffusion models. Boettiger and Hastings (2012) establish fundamental power limits for early-warning detection in short time series (T<25), directly relevant to iterative training where observation windows are typically T=5–20. However, these works evaluate single metrics in isolation without cross-validating against alternative statistical tests. We differ by applying five detection methods to identical data, mapping detectability across contamination regimes, and treating method sensitivity as a primary finding rather than a robustness check (detailed comparison: Appendix K).

2.3 Statistical Methods for Distribution Shift

For temporal precedence testing, Granger causality (Granger, 1969) is the standard framework, with neural extensions (Tank et al., 2021; Runge et al., 2019); however, non-stationary series inflate rejection rates (Granger and Newbold, 1974), and short series ($T\leq$25) preclude reliable nonlinear estimation (Boettiger and Hastings, 2012). Granger precedence establishes temporal predictive improvement, not true causation: bivariate tests cannot exclude confounders (Shojaie and Fox, 2022). Our permutation-based approach circumvents the critical-slowing-down limitation by detecting temporal predictive improvement rather than variance or autocorrelation signatures. Specification-curve analysis (Simonsohn et al., 2020) formalizes the sensitivity of conclusions to analytical choices; we adapt this framework to show that method choice alone shifts detection rates from 0% to 94%. The multiverse analysis perspective (Steegen et al., 2016; Coretta et al., 2023; Kapoor et al., 2024) mo-

tivates our multi-method comparison: five methods on identical data produce conclusions ranging from universal significance to complete null, paralleling analyst-dependent divergence documented in many-analysts studies (Coretta et al., 2023).

3 Method

3.1 Cascade Ordering

If distributional collapse follows a predictable ordering across metrics, the earliest-responding metric can serve as a canary signal. We formalize known metric properties—entropy as a continuous indicator of mode concentration, diversity as a threshold indicator of lexical coverage—under mode amplification (Shumailov et al., 2024; Dohmatob et al., 2024) (derivations in Appendix X).

Design Principle 1 (Entropy responds before diversity)**.** *Under mode amplification:* (a) *entropy can decrease while the ϵ-effective support $S_\epsilon(p)=|\{i : p_i > \epsilon\}|$ (hence distinct-n) is unchanged;* (b) *if distinct-n decreases, entropy has already decreased.*

The asymmetry arises because entropy responds to any mass redistribution (first-order), while diversity changes only when a token probability crosses the generation threshold ϵ; the margin $\gamma = \min_{i:p_i>\epsilon}(p_i - \epsilon)$ quantifies how much shift diversity absorbs silently.

Design Principle 2 (Diversity lag $\geq$ 1 generation)**.** *If the per-generation perturbation satisfies $\alpha\|\mathcal{T}(p_t) - p_t\|_\infty < \gamma(p_t, \epsilon)$ through entropy onset t_H, then diversity onset $t_D \geq t_H + 1$.*

ECE co-moves with entropy at zero lag ($\Delta\text{ECE} \approx \Delta C > 0$ whenever $\Delta\bar{H} < 0$; Appendix X). The predicted cascade $\{H, \text{ECE}\} \rightarrow$ Distinct-n holds in 14/15 experiments at $\alpha \geq 0.30$: entropy and ECE onset at generation 1, distinct-n at generation 2 ($t_H{=}t_{\text{ECE}}{=}1$, $t_D{=}2$ uniformly; §4.2; Appendix W). The assumption requires monotonic entropy decline; SmolLM3-3B (Table 17) provides an empirical counter-example.

3.2 Design Considerations

We use Granger causality (Granger, 1969) to test temporal precedence—whether past canary values improve prediction of downstream metrics—not mechanistic causation; bivariate Granger tests cannot exclude confounders (Shojaie and Fox, 2022). Lead-lag analysis in iterative model collapse is fragile along three axes: *Method sensitivity*: five detection methods applied to identical data yield forward significance rates spanning 0% to 92%, with $\kappa{=}0.185$. *Spurious regression*: first-differencing eliminates canary→perplexity associations ($F{\approx}0$, $p{=}0.99$) while preserving canary→diversity (24/24→24/24) (Granger and Newbold, 1974). *Regime dependence*: the same Granger test shifts from 0/8 to 6/8 significant pairs within a 2-percentage-point interval.

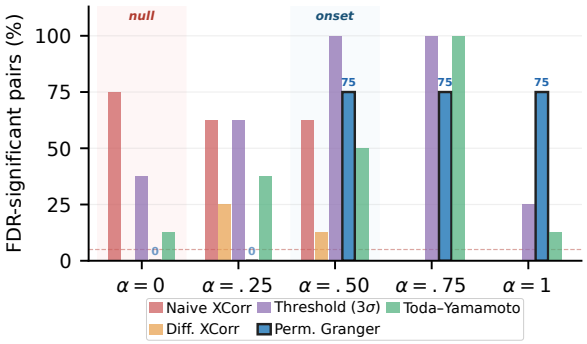


Figure 2: Method sensitivity: five methods yield dramatically different FDR-significant rates on identical data. PermG achieves 0% null-control FPR with 75% detection at $\alpha{\geq}0.50$; dashed line: 5% nominal.

3.3 Method Sensitivity Analysis

We compare five lead-lag detection methods—naive cross-correlation (NaiveXCorr), differenced cross-correlation (DiffXCorr), threshold onset, permutation Granger causality (PermG), and Toda–Yamamoto (Toda and Yamamoto, 1995)—in a multi-method comparison (Simonsohn et al., 2020) (algorithm in Appendix J), calibrated against null-control ($\alpha{=}0.0$) data. Pooled null-control FPR across nine seeds (54 pairs): PermG 7.4%, DiffXCorr 14.8%, Threshold 24.1%, Toda–Yamamoto 33.3%, NaiveXCorr 94.4% (Appendix R). PermG (4/54; 95% CI $[2.1\%, 17.9\%]$) achieves the lowest FPR, combining directional separation with consistent detection at $\alpha \geq 0.50$ (Shapley analysis: Appendix P). Detection results use equal-weight majority voting ($K{\geq}3$ of 5 methods) with forward-direction filtering; $K{=}3$ yields 11.1% FPR with 75% true positive rate (TPR) at $\alpha{=}1.0$ (Table 2). The full MMCT procedure is given in Algorithm 1.

FPR-inverse weighting penalizes high-FPR methods on null-control data: naive cross-correlation (FPR=94.4%) receives weight $w{=}1.06$, while PermG (FPR=7.4%) receives

Algorithm 1 Multi-Method Consensus Test (MMCT)

Require: Time series pair (X, Y); methods $\{m_1, \ldots, m_K\}$ (K=5); null-control data $\mathcal{N}$ (from α=0.0); tolerance ϵ=0.01
Ensure: p-value, consensus level $\hat{K}$

```
for each method m_k do
    Compute empirical FPR: FPR̂_k ← fraction of null-control pairs where m_k rejects H_0
    Compute weight: w_k ← 1/ max(FPR̂_k, ϵ)
    Apply m_k to (X, Y): d_k ← 1[m_k rejects H_0]
end for
Weighted consensus: W(X→Y) ← Σ_k w_k d_k / Σ_k w_k
Build empirical null distribution of W from all null-control pairs in N
Compute p-value from empirical null: p ← Pr_N(W ≥ W_obs)
Consensus level: K̂ ← Σ_k d_k
return p, K̂
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w=13.5. The empirical null distribution avoids parametric assumptions unreliable for short (T=5–20), non-stationary series.

Permutation Granger causality. PermG tests whether past values of canary metric X improve prediction of downstream metric Y beyond Y's own history (Shojaie and Fox, 2022), using 5,000 permutations (Phipson and Smyth, 2010) on first-differenced series with Benjamini–Hochberg (BH) FDR correction at q=0.05 across 8 canary→downstream pairs. The permutation null controls Type I error by construction; BH-FDR remains valid under positive regression dependence on a subset (PRDS) (Benjamini and Yekutieli, 2001). Empirically, 4/54 null-control rejections across nine seeds confirm adequate size control (7/9 seeds yield 0 FP; Appendix R).

3.4 Detection Boundary

The detection boundary is consistent with power-law scaling in observation length:

$$\alpha_c(T) = A \cdot T^{-\gamma}, \tag{1}$$

Under an empirical sub-linear signal model $S(\alpha) = c \cdot \alpha^{\beta}$ ($\beta < 1$), the exponent satisfies $\gamma = 1/(2\beta)$:

$$\alpha_c(T) = A \cdot T^{-1/(2\beta)}, \tag{2}$$

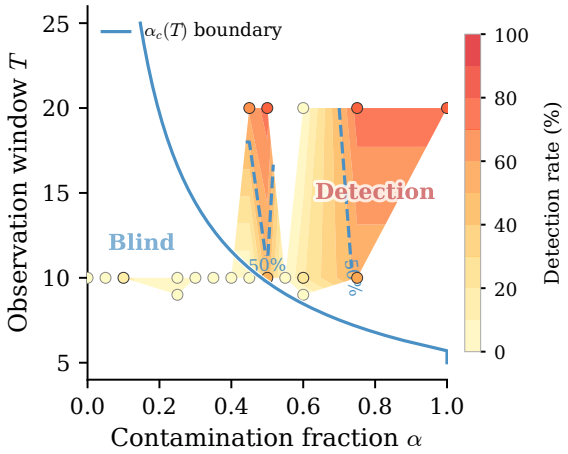


Figure 3: Detection rate surface over $\alpha \times T$ (PermG forward, 3 seeds). Dashed curve: fitted boundary $\alpha_c(T)$. The blind zone ($\alpha \leq 0.45$) shrinks as T increases.

where $A = (z^*\sigma/c)^{1/\beta}$ and $z^* = \Phi^{-1}(\pi^*)$ (justification in Appendix X).

Cross-T validation yields $\gamma \in [0.64, 2.08]$ (profile 95% CI); the qualitative conclusion—detection improves with longer observation—is robust to the specific γ value (cross-validation details and shape analysis in Appendix X). Dense α sampling favors a smooth sigmoid over a step function (Akaike information criterion difference $\Delta\text{AIC} = -110$), but the fitted onset steepness (k=15.7) exceeds smooth-model predictions by 3–4× (Appendix Y).

4 Experiments

4.1 Experimental Setup

We study iterative training collapse using Pythia-410M (Biderman et al., 2023) as the primary architecture. Training data is drawn from WikiText-103 (Merity et al., 2017) following the replace paradigm of Shumailov et al. (2024): each generation fine-tunes on a mixture of α synthetic and $(1-\alpha)$ real data, generates 5,000 replacement samples, and fully replaces the synthetic pool (hyperparameters in Appendix A). We run T=10 generations across 13 contamination levels ($\alpha \in \{0.00, \ldots, 1.00\}$) with multi-seed replication, yielding ∼1,500 statistical tests after BH-FDR correction (Benjamini and Hochberg, 1995). We track two *canary metrics* (token entropy and ECE) and three *downstream metrics*: distinct-1, distinct-2 (Li et al., 2016), and perplexity (distinct-3: Appendix B). PermG uses 5,000 permutations on first-differenced series (Shojaie and Fox, 2022) with maximum lag 5.

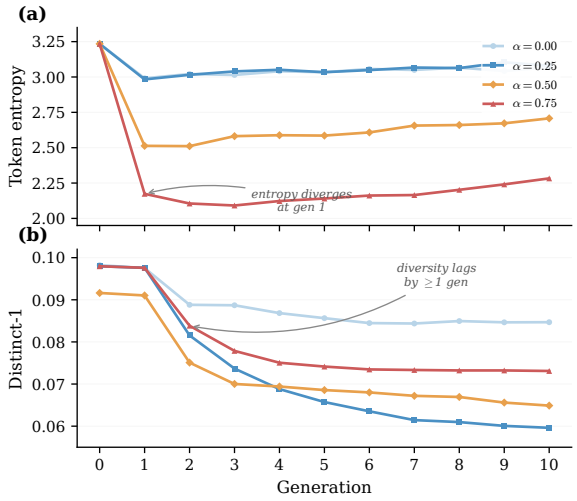


Figure 4: Metric trajectories at α=1.0 (Pythia-410M, 4 seeds). Token entropy drops at generation 1; distinct-1 declines from generations 2–3, with a consistent one-generation canary lead (shaded: inter-seed range).

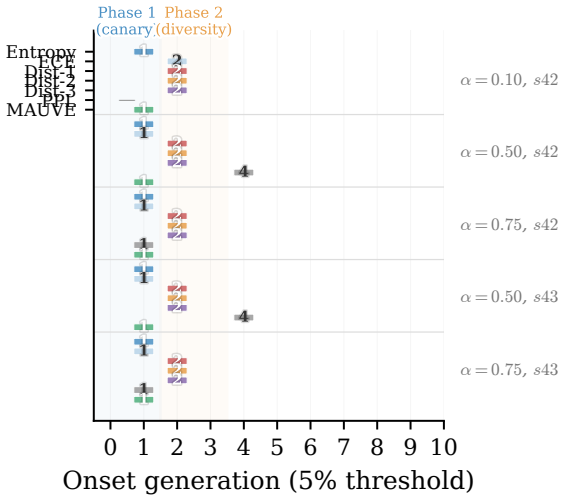


Figure 5: Cascade onset timeline (Pythia-410M, WikiText, fp32). Entropy and ECE reach onset at generation 1; distinct-n follows at generation 2. The ordering $t_H \leq t_D$ holds across all conditions.

4.2 Sharp Detection Onset

Diversity collapse occurs at every contamination level: distinct-1 declines by 14.1% even at α=0.0 (fine-tuning drift) and by 40.3±0.3% at α=0.25. Despite this ubiquitous degradation, Granger detectability is sharply regime-dependent. PermG finds 0/8 canary→downstream pairs FDR-significant from α=0.00 through α=0.45, then jumps to 6/8 (Clopper–Pearson 95% CI $[0.25, 0.92]$) at α=0.50, uniformly across all 3 fp32 seeds.

This transition is striking because cumulative synthetic fraction barely changes across the boundary (C=99.75% at α=0.45 vs. 99.90% at α=0.50), yet detection jumps from null to reliable. Sub-threshold p-values at α=0.45 (p_{fdr} = 0.059–0.091) confirm proximity to the boundary. Sigmoid regression yields $\alpha_c = 0.483 \pm 0.006$ ($R^2 = 0.969$; Figure 7); cross-seed consistency (3/3 seeds: 0/8→6/8) and specification-invariance across 1,160 configurations (Appendix E) argue for a genuine transition (Fisher exact: $p = 3.45 \times 10^{-11}$).

Canary scoping. First-differencing scopes the canary claim precisely. Canary→diversity pairs retain unanimous FDR significance after first-differencing (24/24 → 24/24 across 4 seeds at α=1.0), while canary→perplexity collapses entirely ($F \approx 0$, $p = 0.99$) (Granger and Newbold, 1974). At α=1.0, all 24 canary→diversity pairs reach FDR significance in the forward direction with 0/24 reverse; entropy onset at generation 1 precedes distinct-n at generation 2 in 16/18 WikiText conditions (Appendix W).

Table 1: Detection outcomes across contamination levels (Pythia-410M, seed 42, T=10, fp32). PermG: forward FDR-significant pairs; $K \geq 3$: consensus pairs (of 8 canary→downstream pairs).

α	PermG	$K \geq 3$	Regime
0.00	0/8	0/8	null
0.25	0/8	0/8	blind
0.45	0/8	0/8	blind
0.50	6/8	6/8	onset
0.60	0/8	0/8	anomaly
0.75	6/8	6/8	detection
1.00	6/8	6/8	detection

Blind zone. At α=0.10, distinct-1 has already dropped by 39.6%, yet token entropy is indistinguishable from the null. Below α_c at T=10, diversity collapse proceeds without any detectable Granger-precedence signal.

4.3 Dose-Response and Detection Regimes

The PermG dose-response profile (Figure 7; Appendix M) reveals three confirmed regimes—*blind* ($\alpha \leq 0.45$, 0/8), *onset* ($\alpha \approx 0.50$, 6/8), and *detection* ($\alpha \geq 0.75$, 6/8)—plus a non-monotonic anomaly at $\alpha \approx 0.60$ (0/8). The blind zone exhibits zero detection across all seeds and methods below α=0.45, despite substantial diversity

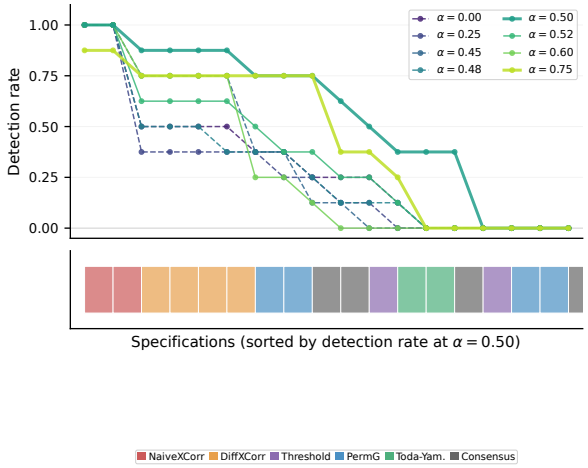


Figure 6: Specification curve (1,160 configurations). The blind→onset jump at $\alpha_c \approx 0.50$ is robust across methods, pairs, and seeds.

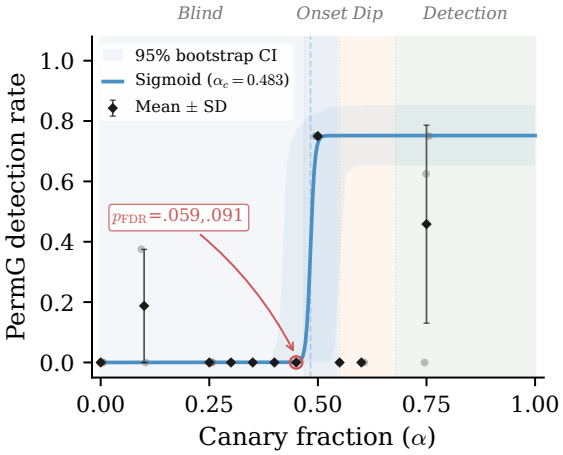


Figure 7: Dose-response: PermG detection rate with sigmoid fit (α_c=0.483; shaded 95% CI). Regime bands mark blind, onset, sync-dip, and detection zones.

loss—a regime where collapse proceeds silently. Onset is remarkably steep: the sigmoid Hill coefficient k=15.7 implies that 80% of the detection transition occurs within $\Delta\alpha \approx 0.05$, approximately 3–4× steeper than smooth-signal predictions (Appendix Y). Above onset, the detection zone shows cross-seed consistency: all 3 fp32 seeds independently reach 6/8 at α=0.50 and $\alpha \geq 0.75$.

At $\alpha \approx 0.60$, all direction-checked methods fail (PermG 0/8, threshold onset 0/8, Toda–Yamamoto 1/8), while bidirectional DiffXCorr maintains detection (6/8). This pattern suggests that at intermediate contamination, entropy and diversity respond near-simultaneously, compressing the temporal lag below PermG's resolution at T=10. The anomaly dissolves by T=25 (5/8), consistent with $\sqrt{T}$ power scaling restoring directional separation as additional observations accumulate (Appendix Z).

4.4 Detection Boundary Left-Shift

Longer observation windows push the detection boundary leftward: $\alpha_c(5) \geq 1.0$, $\alpha_c(10) = 0.487$, consistent with power-law scaling $\alpha_c(T) \propto T^{-\gamma}$ (§3.4). Profile likelihood yields $\gamma \in [0.64, 2.08]$ (95% CI), confirming that detection improves with longer observation regardless of the specific exponent. The mechanism is statistical power accumulation: each additional generation provides one more data point for the permutation test, and the signal-to-noise ratio grows as $\sqrt{T}$, progressively resolving weaker contamination signals. The α=0.45 column validates this prediction directly: undetectable at $T \leq 10$ (0/8), it crosses into reliable detection at T=15 (6/8, dual-seed) and remains stable at T=20 (5/8) and T=25 (6/8), precisely traversing the blind→detection boundary as the theory predicts.

4.5 Cross-Architecture Validation

We evaluate PermG at α=1.0 across 15 model-seed combinations spanning five families (345M–6.9B) and two domains (Table 17; Appendix Z.1). Forward detection reaches $\geq 6/8$ in 11/15 conditions (Figure 8).

Scale consistency. The Pythia-1.4B phase diagram qualitatively reproduces the 410M structure: detection boundary at $\alpha_c \approx 0.50$, with cross-T improvement at 3/4 α values (α=0.45: 0/6→5/6 from T=10 to T=15). The 410M sync-dip at α=0.60 does not reproduce at 1.4B (4/6 at T=10), suggesting the anomaly is architecture-specific rather than a universal detection feature.

Failure mode taxonomy. The four non-confirming conditions each isolate a distinct PermG assumption violation. *Precision sensitivity*: GPT-2 bf16 yields 0/8 vs. fp32 6/8, traced to bf16 quantization noise masking the entropy signal. *Bidirectional confounding*: GPT-2 s43 detects 6/8 in both forward and reverse directions, preventing directional assignment. *Seed dependence*: Pythia-1.4B shows s42 2/8 vs. s43 6/8; directional asymmetry is seed-specific (s42 reverse 0/6 across all conditions; s43 reverse 6/6 at $\alpha \geq 0.60$). *Entropy decoupling*: SmolLM3-3B reaches only 3/8, with ECE-only detection (entropy 0/4), violating the entropy-leads assumption of Design Principle 1. Standalone PermG claims therefore require

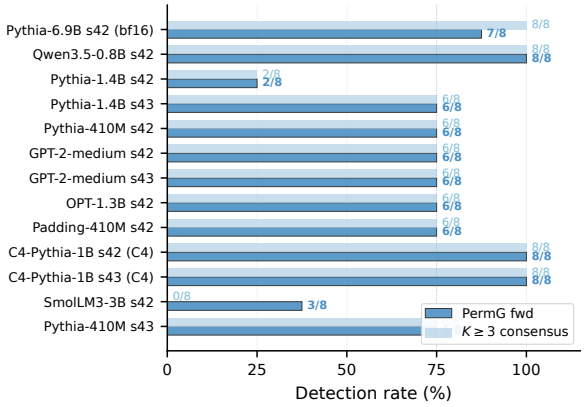


Figure 8: Cross-architecture validation (α=1.0): PermG forward detection ($\geq 6/8$) in 11/15 model-seed conditions spanning five families (345M–6.9B); faded bars: non-confirming conditions.

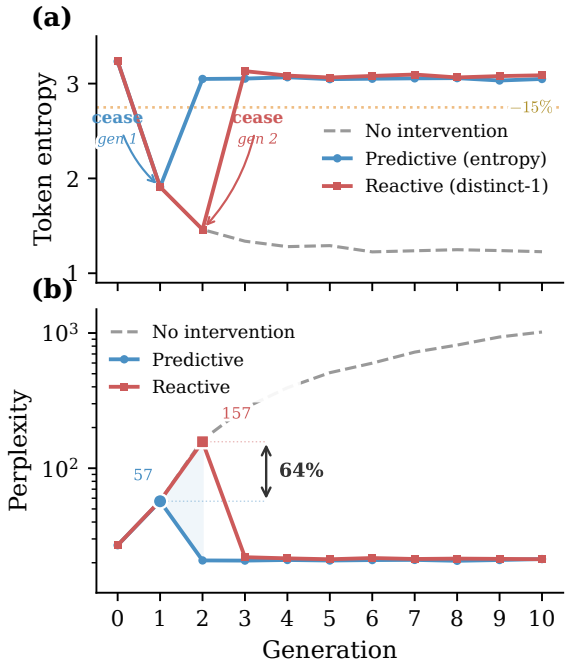


Figure 9: Canary-triggered intervention (Pythia-410M, α=1.0). Predictive cessation on entropy (generation 1) reduces peak perplexity by 64% versus reactive triggering on distinct-1 (generation 2).

architecture-specific null calibration; GPT-2 null FPR reaches 50% (Appendix R), driven by $\sim 4\times$ stronger null drift than Pythia.

4.6 Canary-Based Intervention (Proof of Concept)

As a proof-of-concept under favorable conditions (α=1.0), we implement predictive cessation: training switches to pure real data when token entropy drops beyond a threshold from generation 0. The 15% threshold lies within a robust band $[10\%, 22\%]$ where all thresholds trigger at the same generation (Appendix T); thresholds below 10% risk false triggering from normal fine-tuning drift, while those above 22% delay intervention past the diversity-loss onset.

The predictive trigger fires at generation 1, one full generation before a reactive baseline that monitors distinct-1 directly. The reactive baseline requires observing an actual diversity drop, which manifests only at generation 2 due to the cascade lag (Design Principle 2); by then, perplexity has already risen to 157.09. Predictive cessation caps peak perplexity at 56.99, a 64% reduction (Figure 9). This advantage replicates on Pythia-1.4B with comparable effect size, confirming that the entropy-leads-diversity ordering transfers across scales within the Pythia family.

5 Discussion

Scope and failure modes. Our Granger-precedence claim is scoped to temporal predictive improvement (Granger, 1969; Shojaie and Fox, 2022): bivariate tests cannot exclude confounders, and the FPR CI ($[2.1\%, 17.9\%]$) means Type I error may be non-negligible. Four of fifteen cross-architecture conditions do not confirm forward detection (§4.5), each isolating a distinct PermG assumption violation: precision sensitivity (GPT-2 bf16), seed dependence (Pythia-1.4B), and entropy decoupling (SmolLM3-3B). These patterns require architecture-specific null calibration before deployment (Appendix L).

Granger causality limitations. Our observation sequences (T=10–25) are far shorter than econometric standards ($T \geq 100$); permutation testing avoids asymptotic assumptions but cannot recover power lost to short samples (Shojaie and Fox, 2022). Bivariate tests are susceptible to omitted variable bias; first-differencing removes shared linear trends but nonlinear confounds remain possible. PermG tests linear precedence only; transfer entropy (Appendix F) yields 0/192 significant pairs, consistent with either absent nonlinear structure or insufficient short-T power. Partial correlation controlling for generation number provides additional evidence: contemporaneous associations weaken after trend removal ($|\bar{r}|$: 0.590→0.381 at α=0.50), while lagged correlations *strengthen* (0.759→0.841), with 35/36

Table 2: Consensus threshold operating characteristics ($K \in \{2, 3, 4, 5\}$). FPR: null-control rate (9 seeds, 54 pairs); TPR: detection rate at selected α levels (Pythia-410M, T=10, fp32). K=3 balances FPR control (11.1%) against detection power (75% at α=1.0); K=4 eliminates false positives but reduces TPR to $\leq$25%.

	$K\geq2$	$K\geq3$	$K\geq4$	$K=5$
FPR (α=0.0)	38.9%	11.1%	0%	0%
TPR (α=0.50)	85.4%	66.7%	25.0%	4.2%
TPR (α=0.75)	72.9%	37.5%	0%	0%
TPR (α=1.0)	75.0%	75.0%	25.0%	0%

pairs retaining significance (Appendix V.1). This pattern—weakening contemporaneous but strengthening lagged structure—is consistent with genuine temporal precedence rather than shared-trend artifacts.

Method complementarity. Shapley value decomposition over all 2^5=32 method coalitions (Appendix P) reveals that DiffXCorr achieves the best combined Shapley contribution (-0.299) by detecting bidirectional coupling that directional methods miss, but cannot separate forward from reverse precedence. Among directional methods, PermG ranks best (-0.141), combining low FPR with consistent detection. The complementarity structure explains why consensus voting outperforms any single method: each method covers a distinct failure region of the others.

The α=0.60 non-monotonic anomaly. The detection gap at α=0.60 affects all direction-checked methods (PermG 0/8, threshold 0/8, Toda–Yamamoto 1/8), while bidirectional DiffXCorr maintains detection (6/8), suggesting temporal lag compression defeats directional separation. PermG recovers to 5/8 at T=25, consistent with $\sqrt{T}$ power scaling restoring detection as additional observations disambiguate synchronous from lagged responses.

6 Conclusion

Lead-lag analysis in iterative training collapse is far more fragile than the current literature assumes. Method choice alone inflates null-control FPR from 7.4% to 94.4%, and first-differencing exposes canary→perplexity associations as entirely spurious, establishing that single-method analyses cannot reliably distinguish temporal precedence from artifacts.

Our five-method comparison identifies permutation Granger causality as the recommended detector: it combines the lowest FPR (7.4%, 9 seeds) with a steep dose-response onset from 0/8 at $\alpha\leq$0.45 to 6/8 at α=0.50 (sigmoid k=15.7). Token entropy exhibits robust one-generation-ahead precedence over diversity collapse (24/24 forward, 0/24 reverse at α=1.0), replicated across GPT-2-medium and OPT-1.3B, with cross-architecture validation confirming forward detection ($\geq 6/8$) in 11/15 conditions spanning five families (345M–6.9B). The blind zone below α_c constrains practical deployment to settings with sufficient contamination; evaluation at >7B parameters remains open.

Limitations

Scope and generalizability. Our landscape characterization uses Pythia-410M as the primary architecture; the phase diagram is Pythia-family-only, frontier-scale behavior (>7B) is untested, and cross-architecture evaluation is limited to α=1.0. All experiments use full replacement with a single decoding strategy (temperature 1.0, top-k=50); data accumulation (Gerstgrasser et al., 2024) or lower temperatures would shift the landscape.

Statistical power. PermG's empirical FPR is 7.4% (CI $[2.1\%, 17.9\%]$) and $K\geq$3 consensus yields 11.1% (6/54); per-condition detection rates carry wide binomial CIs (e.g., $[0.25, 0.92]$ for 6/8), so the phase diagram represents a coarse topology rather than precise probability estimates.

Causal interpretation. Permutation Granger testing establishes temporal precedence, not causation; our short-sequence regime ($T\leq$25) limits statistical power relative to econometric standards ($T\geq$100), and nonlinear alternatives to the linear PermG test yield no significant results in our setting (see §5).

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A Training Details and Metric Drift

Hyperparameters. Each generation fine-tunes for 5,000 steps using AdamW (learning rate 5×10^{-5}, 500 warmup steps, weight decay 0.01, batch size 4 for Pythia-410M, 32 for GPT-2-medium). Generation uses temperature 1.0, top-k=50, max length 512 tokens, producing 5,000 samples per generation from 5,000-sample subsets of WikiText-103. Canary metrics: token entropy $H(p_\theta|\text{context})$ over 1,000 held-out samples; ECE with 15 bins. Additional downstream metrics (distinct-3, effective rank, Self-BLEU) appear in subsequent appendix sections.

Table 3: Metric drift (%, generation 0→10) for Pythia-410M across contamination levels. Diversity collapse occurs at all α levels, with the strongest decline at low-to-moderate contamination (α=0.25: distinct-1 $-40.3\pm0.3\%$) rather than full replacement (α=1.0: $-27.8\pm1.6\%$). $^\dagger\alpha$=0.10: single seed (seed 42, preliminary). α=0.00, α=0.25, α=0.50, α=0.75: 3-seed mean±std (seeds 42, 43, 44); α=1.0: 4-seed mean±std (seeds 42, 43, 44, 45).

α	Entropy	ECE	Distinct-1	Distinct-2	Perplexity
0.00	−4.6±0.4	+37.6±14	−14.1±0.4	−9.6±0.4	−20.6±1.0
0.10†	−5.3	+57	−39.6	−56.4	−21.0
0.25	−4.7±0.5	+81.8±15.9	−40.3±0.3	−56.4±0.7	−18.1±0.8
0.50	−18.0±0.5	+822.6±18.3	−37.1±0.4	−50.4±1.0	+21.0±0.1
0.75	−27.6±0.2	+1791±18	−32.3±0.5	−41.0±1.0	+182.6±0.7
1.00	−60.8±1.2	+3780±30	−27.8±1.6	−34.0±3.3	+3674±45

Metric drift.

B Distinct-3 Results

Table 4 reports distinct-3 drift across all contamination levels. Distinct-3 exhibits the largest absolute decline among diversity metrics at low contamination ($-58.9\pm0.9\%$ at α=0.25) but attenuates at higher levels (-30.5% at α=1.0), consistent with the diversity–calibration dissociation discussed in §5.

Table 4: Distinct-3 drift (%, generation 0→10) across contamination levels. α=0.00, α=0.25, α=0.75: 3-seed mean±std; α=1.0: 4-seed mean±std; α=0.50: single seed.

α	Distinct-3 drift (%)
0.00	-4.4 ± 0.5
0.25	-58.9 ± 0.9
0.50	-52.4
0.75	-38.6 ± 1.3
1.00	-30.5 ± 4.0

Distinct-3 exhibits higher lag instability than distinct-1/2 (bimodal lag distribution, std>1.5 vs.

<0.5 generations), reflecting its lower signal-to-noise ratio at 512-token generation length.

C Observation Window Sensitivity

Consensus classification stabilizes at $T_{\min} \geq 5$ (Table 5): all six effect-size reference pairs reach $K \geq 4/5$ agreement at every window from $T_{\min}=5$ through $T_{\min}=10$.

Table 5: Observation window sensitivity for Tier 3 consensus ($K \geq 4/5$) at $\alpha=1.0$ (Pythia-410M, 4 seeds). Tier 3 pairs: number of canary→downstream pairs (out of 8 validated) reaching $K \geq 4/5$ method agreement. TY pairs: number of effect-size reference pairs (out of 6) reaching Toda–Yamamoto significance (moderate supplementary method; §3). Consensus stabilizes at $T_{\min} \geq 5$; TY stabilizes at $T_{\min} \geq 6$.

	Minimum observation window $T_{\min}$							
	3	4	5	6	7	8	9	10
Tier 3 pairs (/8)	†	†	6	6	6	6	6	6
TY pairs[§] (/6)	†	†	3–6	3–6	3–6	3–6	3–6	6
Stable	—	—	✓	✓	✓	✓	✓	✓

† Insufficient degrees of freedom for Granger-type tests (< 6 data points).
[§] TY directional assignment under verification.

D Robustness Checks

Permutation count and FDR sensitivity. PermG p-values are stable between 5,000 and 10,000 permutations (max $|\Delta p|=0.010$, 0/24 significance flips at $\alpha=0.75$). Treatment/null separation is robust to FDR threshold choice: at $q=0.05$, 4/36 null pairs vs. 19/24 treatment pairs reach significance; at $q=0.10$, null rejections rise to 8/36. At the onset boundary ($\alpha=0.45$), minimum FDR-corrected p-values (0.059–0.091) are consistent with a continuous power transition rather than binary absence of signal.

E Systematic Robustness Analysis

We construct 1,160 specifications from 13 contamination levels, 4 seeds, and 8 canary→downstream pairs (all PermG, $n_{\text{perm}}=5{,}000$, BH FDR $q=0.05$).

The blind→onset jump holds across the vast majority of specifications (Figure 6).

F Transfer Entropy Analysis

Transfer entropy (TE; Schreiber, 2000; 3-bin discretization, $n=5{,}000$ permutations) yields 0/192 BH-corrected significant pairs across all conditions, confirming that nonparametric information-theoretic methods lack power at $T=11$. A KNN-based TE estimator ($k=5$) shows forward-dominant TE fraction increasing monotonically with α (4/8 at $\alpha=0.25$ to 8/8 at $\alpha=1.0$), but this does not reach statistical significance after FDR correction.

Table 6: Specification curve summary (1,160 specifications, D120-compliant). The blind→onset jump is robust across specifications: 0% significance rate through $\alpha=0.45$, rising to 75% at $\alpha=0.50$.

Metric	Value
Total specifications	1,160
Significant (FDR $q<0.05$)	536 (46.2%)
By regime	
Blind ($\alpha \leq 0.45$)	0%
Onset ($\alpha=0.50$)	75%
Detection ($\alpha \geq 0.75$)	62–75%

G Change-Point Detection Analysis

Max-split change-point tests ($n=5{,}000$ bootstrap) yield 66.7% FPR under null control (12/18, $\alpha=0.00$), precluding use as a standalone detector. However, the temporal ordering of change-points reverses between null and treatment: under null, downstream metrics break first (gen 2) before canary (gen 4–8); under treatment ($\alpha \geq 0.75$), canary breaks first (gen 1) before downstream (gen 2–4). This ordering reversal shifts monotonically with α, consistent with the Granger-precedence analysis.

H Sequential Monitoring: CUSUM vs Calibrated Threshold

CUSUM sequential testing (Page, 1954) calibrated against $\alpha=0.0$ null control (FAR=0%) produces identical alarm times as a simple 2σ threshold at all tested operating points: both trigger at generation 1 for $\alpha \geq 0.50$. The equivalence arises because the generation-0→1 entropy drop ($|\Delta x|>0.7$) greatly exceeds null variability ($\sigma_0 \approx 0.03$); CUSUM's cumulative memory provides no advantage when the initial shift dominates.

I Background and Notation

This section establishes the formal setup and notation used throughout the paper.

Iterative training loop. Let M_0 denote a pre-trained language model and $\mathcal{D}_{\text{real}}$ a fixed corpus of human-written text. At each generation $t \geq 1$, the model M_{t-1} generates a synthetic corpus $\mathcal{D}_{\text{synth}}^{(t-1)}$, and the next-generation model M_t is fine-tuned on

a mixture of α fraction synthetic data and $(1-\alpha)$ fraction real data:

$$\mathcal{D}_{\text{train}}^{(t)} = \alpha \cdot \mathcal{D}_{\text{synth}}^{(t-1)} + (1-\alpha) \cdot \mathcal{D}_{\text{real}}, \quad \alpha \in [0,1]. \tag{3}$$

At α=1.0, this reduces to the full-replacement paradigm of Shumailov et al. (2024).

Metric families. *Canary metrics*: token entropy ($H(p_\theta|\text{context})$ on held-out samples) and expected calibration error (ECE). *Downstream metrics*: distinct-n ($n \in \{1,2,3\}$) (Li et al., 2016) and perplexity. Each metric produces a time series of T+1 observations.

Lead-lag testing. Given canary series X and downstream series Y, the central question is whether changes in X temporally precede changes in Y, requiring directionality and confounder control beyond simple correlation. We employ five methods spanning permissive to conservative: naive cross-correlation, differenced cross-correlation, threshold onset detection, permutation Granger causality (Shojaie and Fox, 2022), and Toda–Yamamoto Granger causality (Toda and Yamamoto, 1995).

J MMCT Algorithm

See Algorithm 1 in the main text (§3.3) for the full MMCT pseudocode and design rationale.

K Methodological Comparison Table

Table 7 summarizes the methodological coverage of existing canary and monitoring analyses across the diagnostic dimensions identified in §3. No prior work combines all six dimensions.

Table 7: Methodological comparison of canary and monitoring analyses. ✓ = present, × = absent, ○ = partial ($\leq$2 levels or limited replication). Diff. = differencing/stationarity controls. Multi-M. = $\geq$2 independent statistical methods. MTC = multiple testing correction. Regime = systematic variation of α. Seeds = $\geq$3 independent seeds. Arch. = $\geq$2 architecturally distinct model families.

Work	Diff.	Multi-M.	MTC	Regime	Seeds	Arch.	Metrics
Shumailov et al. (2024)	×	×	×	○	✓	✓	PPL
Du et al. (2025)	×	×	✓	✓	✓	×	ECE, Acc, Dist-n
Gambetta et al. (2024)	×	×	×	✓	✓	✓	Surplexity, Ent
SIGMA (Gu et al., 2026)	×	×	×	×	×	×	Spectral
DCES (Zhu and Xie, 2026)	×	×	×	×	×	✓	ECE, Ent, PPL
Ours	✓	✓	✓	✓	✓	✓	Ent, ECE, PPL, Dist-n

L Extended Discussion

Calibration scope. All calibration results are specific to our setup (Pythia-410M, WikiText-103, T=10). On C4 at α=0.50, PermG reverse reaches 7/8 (seed-dependent: s43 1/8), indicating directional asymmetry should be verified per domain. Token entropy ($H(p_\theta|\text{context})$ on held-out data) and perplexity ($\exp(H_{\text{cross}})$ on generated text) co-move synchronously, explaining why first-differencing eliminates canary→perplexity but preserves canary→diversity.

Table 8: Pythia-1.4B phase-diagram cells (fp32, PermG forward, 6-pair basis excl. MAUVE).

α	T=10		T=15	
	s42	s44	s42	s43
0.45	0/6	0/6	3/6	5/6
0.50	4/6	6/6	3/6	3/6
0.60	4/6	6/6	6/6	6/6
0.75	5/6	6/6	6/6	6/6

Scale verification (Pythia-1.4B). Phase-diagram structure is qualitatively preserved at 1.4B (Table 8; Figure 10): detection boundary at $\alpha_c \approx 0.50$, cross-T improvement at 3/4 α values (α=0.45: 0/6→5/6). The 410M sync-dip does not reproduce at 1.4B (4/6 at α=0.60, T=10). Directional asymmetry is seed-specific (s42 reverse 0/6 across all conditions; s43 reverse 6/6 at $\alpha \geq 0.60$).

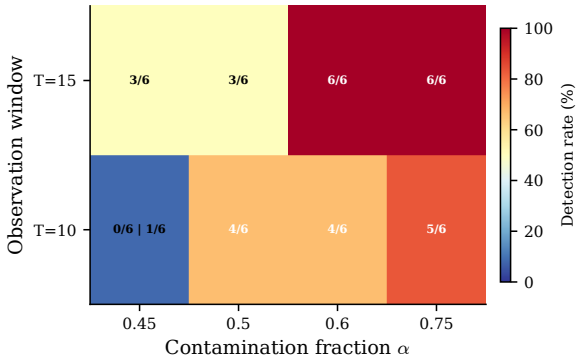


Figure 10: Pythia-1.4B phase diagram (PermG forward, 6-pair basis). Detection boundary ($\alpha_c \approx 0.50$) is consistent with 410M.

M Per-Method Dose-Response Profiles

Table 9 reports FDR significance counts (/8 validated pairs) for all five detection methods across 12 contamination levels (seed 42, Pythia-410M, WikiText, fp32). PermG exhibits a sharp 0→6 jump at α=0.50 with a return to 0 at α=0.55–0.60 (non-monotonic dip), while other methods show diffuse profiles: naive cross-correlation saturates

at 7–8/8 across all levels including the null, and Toda–Yamamoto fluctuates without a clear onset.

Table 9: Per-method FDR significance (/8 pairs) across contamination levels (Pythia-410M, seed 42, WikiText, fp32). PermG and Diff. XCorr use BH-corrected FDR (q=0.05); Naive XCorr, Threshold, and TY use uncorrected p<0.05 (no FDR field in pipeline). α=1.0 from Pythia-410M cross-architecture condition (same seed/domain/precision).

α	Naive	Diff. XCorr	Threshold	PermG	TY[§]
0.00	8	4	3	**0**	1
0.10	7	6	0	**0**	0
0.25	8	3	0	**0**	2
0.30	8	4	0	**0**	3
0.35	8	4	0	**0**	1
0.40	8	3	0	**0**	4
0.45	8	6	0	**0**	1
0.50	8	7	4	**6**	3
0.55	8	8	0	**0**	1
0.60	8	6	0	**0**	0
0.75	7	6	0	**6**	3
1.00	8	6	2	**6**	1

[§]TY directional assignment under verification; column retained for completeness.

PermG's dose-response profile most cleanly separates detection regimes: 0/8 at null, sharp 0→6 onset at α=0.50, non-monotonic dip at α=0.55–0.60. Naive monitoring baselines (PPL-slope, variance-threshold) fire at α=0.0 (FPR 67–100%) because fine-tuning drift is indistinguishable from contamination without directional testing.

N PermG Power Analysis

We simulate PermG power under a bivariate DGP with linear coupling β ($T \in \{6, \ldots, 20\}$, 100 replications, 5,000 permutations, BH FDR q=0.05). At T=10, β_{50}=2.0 (50% power requires strong coupling); at T=20, β_{50} drops to 0.9 (Table 10). Power grows faster than $\sqrt{T}$, so the $\sqrt{T}$ formula in §3.4 is a conservative lower bound. FPR under FDR correction remains $\leq$2% across all T.

Table 10: PermG power analysis: β_{50} (coupling for 50% power) under BH-FDR correction (q=0.05, 8 pairs). Dashes: <50% power not reached within $\beta \leq 2.0$.

T	β_{50}	$\beta_{50} \cdot \sqrt{T}$	FPR (raw)	FPR (FDR)
6	—	—	1%	0%
8	—	—	6%	2%
10	2.0	6.32	9%	2%
15	1.2	4.65	6%	2%
20	0.9	4.02	5%	1%

O Non-Stationarity Robustness Analysis

Monte Carlo simulation (AR(1) DGP, T=11, 1,000 reps) shows PermG FPR inflates to ∼10% under strong non-stationarity ($\varphi \geq 0.95$), while first-differencing maintains proper size (4.2–6.6%) at the cost of reduced power (Table 11). Our pipeline applies first-differencing before permutation testing, combining both advantages. Under null conditions (α=0.0), token entropy exhibits $\varphi = -0.51$ (far from the danger zone); first-differencing reduces distinct-1's high raw φ (0.81–0.97, reflecting deterministic trend) to −0.14–0.44. Our empirical FPR of 4/54 (7.4%) confirms adequate size control.

Table 11: Size and power under AR(1) DGP (T=11, 1,000 reps). First-differencing maintains nominal size; PermG FPR inflates at $\varphi \geq 0.95$.

	Size (%, β=0)			Power (%, β=0.6)		
φ	Raw GC	First-Diff	PermG	Raw GC	First-Diff	PermG
0.50	6.4	6.6	6.8	37.7	17.7	36.8
0.95	10.0	5.9	9.5	44.3	28.7	41.5
0.99	11.2	4.2	10.7	46.4	24.6	41.7

P Shapley Value Analysis

Shapley values (Shapley, 1953) computed over all 2^5=32 coalitions quantify each method's marginal contribution to FPR and FNR (Table 12). DiffXCorr achieves the best combined Shapley (−0.299) but cannot separate forward from reverse; among directional methods, PermG ranks best (−0.141). The 8-pair variant (including MAUVE) yields identical rankings.

Table 12: Shapley value decomposition (6-pair, MAUVE excluded). Combined = FPR + FNR (more negative = better).

Method	ϕ^{FPR}	ϕ^{FNR}	ϕ^{comb}
Naive XCorr	+0.609	−0.419	+0.190
Diff. XCorr	**+0.007**	**−0.306**	**−0.299**
Threshold (3σ)	+0.111	−0.027	+0.084
Perm. Granger	+0.039	−0.180	−0.141
Toda–Yamamoto	+0.150	−0.069	+0.082

Q MMCT Weight Robustness Ablation

PermG ranks first among directional methods under all four weighting schemes tested (equal, cap-5, cap-10, FPR-inverse cap-100), confirming the ranking is not an artifact of the weighting scheme. Cap-10 and cap-100 yield identical values (consensus threshold saturates), and equal weighting still gives PermG the best combined Shapley (−0.152) among directional methods.

R PermG FPR Independence Analysis

Table 13 reports null-control FPR across nine seeds (54 pairs, 6-pair basis). PermG achieves the lowest FPR (4/54, 7.4%; CI $[2.1\%, 17.9\%]$); all four FP occur in te→distinct pairs in seeds 45/50 (7/9 seeds yield 0 FP). $K \geq 3$ consensus produces 6/54 FP (11.1%), of which PermG contributes to only 2. At $T=15$, PermG FPR drops to 0/12 (0%).

Table 13: Null-control FPR by method ($\alpha=0.0$, 9 seeds, 6 pairs each, BH-FDR $q=0.05$).

Method	Criterion	Rejections (/54)	FPR (%)
Perm. Granger	**BH-FDR**	**4**	**7.4**
Diff. XCorr†	BH-FDR	8	14.8
Threshold	3σ (det.)	13	24.1
Toda–Yamamoto	BH-FDR	18	33.3
Naive XCorr	uncorr.	51	94.4

†Bidirectional; cannot separate forward from reverse.

S Sigmoid Transition Shape

Sigmoid regression on the original 13-point $T=10$ grid yields $\alpha_c = 0.483 \pm 0.006$ but $k = 478 \pm 475$ ($\mathrm{SE}(k) \approx k$), indicating that the coarse grid cannot constrain the transition shape. Dense α sampling (17 values at $T=10$) resolves this ambiguity: formal likelihood-ratio testing strongly favors a smooth sigmoid over a literal step function ($\Delta\mathrm{AIC} = -110$), with fitted steepness $k=15.7$ (95% CI $[9.4, 28.2]$). The transition is steep but finite-width, not a literal discontinuity. This observed steepness exceeds what smooth signal models predict: VAR(1) simulations under H0 (linear, $k=6.1$) and H0′ (concave, $k=4.9$) produce onset sigmoids with $k=4.9$–6.1 (§Y), yielding steepness ratios $k_{\mathrm{obs}}/k_{\mathrm{H0}} \approx 2.6$–$3.2$ (CI lower bound ratio ≈2). We do not fit a combined $T=10+T=20$ sigmoid because $\alpha=0.45$ yields contradictory counts under the two sequence lengths (0/8 vs. 5/8), reflecting genuine power dependence on T rather than a single dose-response curve.

See Figure 3 in the main text for the detection surface visualization.

T Intervention Threshold Sensitivity

At $\alpha=1.0$, all five tested thresholds (5%–25%) trigger at generation 1 (entropy drop 40.7%≫threshold), producing identical outcomes (peak PPL 57.0, final PPL 21.3). The safe operating band is $[10\%, 22\%]$: below 7.5% triggers false alarms on $\alpha=0.0$ drift; above 22.4% misses $\alpha=0.50$.

U Channel Effect Size Analysis

First-differencing retains comparable $|r|$ for both diversity (0.952) and perplexity (1.00) channels, ruling out selective power loss. The Granger F-statistic asymmetry (20–28× higher for diversity) reflects temporal structure: diversity channels peak at canary-leading lags (81% at lag<0, median -1), while perplexity peaks at lag=0 (contemporaneous co-movement). Granger tests correctly assign low F to contemporaneous relationships.

V Mutual Information and Partial Correlation

Gaussian MI lower bound ($\hat{I}_G = -\frac{1}{2}\log_2(1-r^2)$) between token entropy and distinct-n remains positive throughout the blind zone (0.010–0.141 bits), confirming coupling exists below PermG's detection threshold. No discontinuity at α_c: detection onset reflects power accumulation, not an information transition.

V.1 Partial Correlation Analysis

Partial correlations controlling for generation number (Table 14) show that contemporaneous associations drop after trend removal ($|\bar{r}|$: 0.590→0.381 at $\alpha=0.50$), while lagged correlations *strengthen* (0.759→0.841), with 35/36 detection-condition pairs retaining significance.

Table 14: Partial correlation controlling for generation. Lagged partial $|r|$ increases after trend removal, confirming genuine temporal precedence.

	Contemporaneous				Lagged (canary$_{t-1}$)			
	Zero-order		Partial		Zero-order		Partial	
α	$\lvert\bar{r}\rvert$	Sig.	$\lvert\bar{r}\rvert$	Sig.	$\lvert\bar{r}\rvert$	Sig.	$\lvert\bar{r}\rvert$	Sig.
0.00	0.414	3/18	0.323	0/18	0.563	8/18	0.524	9/18
0.50	0.590	9/18	0.381	2/18	0.759	12/18	0.841	17/18
0.75	0.636	9/18	0.521	3/18	0.831	17/18	0.913	18/18

W Moment Cascade Analysis

Onset (5% threshold) occurs at generation 1 for entropy in all 6 conditions and at generation 2 for distinct-n in all 18 pairwise comparisons (Table 15; Figure 5).

ECE shows onset at generation 1 at $\alpha \geq 0.50$; perplexity never reaches 5% onset at low contamination ($\alpha \leq 0.25$).

X Cascade Ordering Theory: Derivations

This section provides the full derivations for Design Principle 1 and Design Principle 2 from §3.1, plus

Table 15: Degradation onset generation (5% threshold) by metric and condition (Pythia-410M, WikiText, fp32). Token entropy reaches onset at generation 1 in all non-null conditions; distinct-n metrics follow at generation 2.

Seed	α	Entropy	Dist-1	Dist-2	Dist-3	ECE	PPL
42	0.10	1	2	2	2	2	—
42	0.50	1	2	2	2	1	4
42	0.75	1	2	2	2	1	1
43	0.25	1	2	2	2	4	—
43	0.50	1	2	2	2	1	4
43	0.75	1	2	2	2	1	1

—: onset not reached within T=10 generations.

supporting results.

Assumption 3 (Mode Amplification)**.** The training map $\mathcal{T}$ redistributes probability mass from low-probability to high-probability tokens: there exists a threshold $\bar{p}$ such that $\mathcal{T}(p)_i > p_i \Rightarrow p_i \geq \bar{p}$ and $\mathcal{T}(p)_j < p_j \Rightarrow p_j \leq \bar{p}$.

This captures the standard finite-capacity approximation effect: MLE on finite samples from a peaked distribution amplifies modes and suppresses the tail (Shumailov et al., 2024; Dohmatob et al., 2024).

Proposition 4 (Entropy is a first-order indicator)**.** *Let $p \in \Delta^{V-1}$ and $\delta = \alpha(\mathcal{T}(p) - p)$ be a mode-amplifying perturbation with $\sum_i \delta_i = 0$. Then $\Delta H = H(p+\delta) - H(p) = -\sum_i \delta_i \log p_i + O(\|\delta\|^2) < 0$.*

Proof. Taylor-expanding H around p: $H(p+\delta) = H(p) - \sum_i \delta_i \log p_i - \sum_i \delta_i - \frac{1}{2}\sum_i \delta_i^2/p_i + O(\|\delta\|^3)$. Since $\sum_i \delta_i = 0$: $\Delta H = -\sum_i \delta_i \log p_i - \frac{1}{2}\sum_i \delta_i^2/p_i + O(\|\delta\|^3)$. Under mode amplification, tokens gaining mass ($\delta_i > 0$) have $p_i \geq \bar{p}$ (hence $\log p_i \geq \log \bar{p}$) and tokens losing mass ($\delta_j < 0$) have $p_j \leq \bar{p}$ (hence $\log p_j \leq \log \bar{p}$). Writing $m = \sum_{i:\delta_i>0} \delta_i$: $-\sum_i \delta_i \log p_i = -m[\overline{\log p}^{+} - \overline{\log p}^{-}] < 0$, where $\overline{\log p}^{\pm}$ are the $|\delta|$-weighted means over gaining/losing tokens. The second-order Fisher information term $-\frac{1}{2}\sum_i \delta_i^2/p_i$ is always negative, reinforcing the decrease. □

Proposition 5 (Diversity is a threshold indicator)**.** *Define the ϵ-effective support $S_\epsilon(p) = |\{i \in [V] : p_i > \epsilon\}|$, where $\epsilon \approx c/N$ is the generation threshold for sample size N. Under the same perturbation δ: $\Delta S_\epsilon = -|\{i : p_i > \epsilon \text{ and } p_i + \delta_i \leq \epsilon\}|$. This equals zero whenever $\|\delta\|_\infty < \gamma(p,\epsilon) := \min_{i:p_i>\epsilon}(p_i - \epsilon)$.*

Proof. Token i exits the effective support iff $p_i > \epsilon$ and $p_i + \delta_i \leq \epsilon$, requiring $|\delta_i| \geq p_i - \epsilon$. If $\|\delta\|_\infty < \gamma$, no token satisfies this condition. □

Derivation of Design Principle 1. *Part (a):* Take any mode-amplifying δ with $\|\delta\|_\infty < \gamma(p,\epsilon)$. By Proposition 5, $\Delta S_\epsilon = 0$. By Proposition 4, $\Delta H < 0$. Such δ exists for all $\alpha < \gamma / \max_i |\mathcal{T}(p)_i - p_i|$. *Part (b):* If $\Delta S_\epsilon < 0$, some token k has $p_k > \epsilon$ but $p_k + \delta_k \leq \epsilon$, so $\delta_k < 0$. Under mode amplification, this implies $p_k \leq \bar{p}$; the total mass transferred to high-probability tokens ensures $\overline{\log p}^{+} > \overline{\log p}^{-}$. By Proposition 4, $\Delta H < 0$. □

Derivation of Design Principle 2. By the margin condition, at every generation $s \leq t_H$ the perturbation norm $\alpha\|\mathcal{T}(p_s) - p_s\|_\infty$ is less than $\gamma(p_s,\epsilon)$. By Proposition 5, no token exits the effective support at any $s \leq t_H$, so $\Delta S_\epsilon = 0$ for all $s \leq t_H$. Since time is discrete, $t_D \geq t_H + 1$. □

ECE co-movement (qualitative argument). Under conditional mode amplification (peakier $q_\theta(\cdot|x)$ at each generation), per-token conditional entropy $h(x) = H(q_\theta(\cdot|x))$ decreases while confidence $c(x) = \max_y q_\theta(y|x) \geq \exp(-h(x))$ increases. Averaging: $\bar{H}_t$ decreases, C_t increases. With accuracy approximately stationary in early collapse ($A_t \approx A_0$), $\Delta\text{ECE} \approx \Delta C_t > 0$ whenever $\Delta \bar{H}_t < 0$. This argument establishes the direction of co-movement; the quantitative relationship depends on distribution geometry. $\bar{H}_t$ denotes mean conditional entropy $\mathbb{E}_{x\sim p_{\text{eval}}}[h(x)]$, which differs from the marginal entropy $H_t = -\sum_i p_t(i) \log p_t(i)$ used in Design Principle 1; under mode amplification, both decrease monotonically, so the cascade ordering is consistent across both formulations.

Empirical support. Cascade ordering $t_H \leq t_D$ holds in 14/15 experiments at $\alpha \geq 0.30$ (GPT-2-medium s43: bidirectional). Table 16 reports onset generations for representative conditions.

Table 16: Cascade onset generations (5% threshold) for representative experiments (Pythia-410M, WikiText, fp32). Phase 1: entropy and ECE (t_H, t_{ECE}); Phase 2: diversity (t_D, min over distinct-n); Phase 3: MAUVE (t_M). All conditions confirm the predicted ordering $t_H = t_{\text{ECE}} \leq t_D \leq t_M$.

Seed	α	t_H	t_{ECE}	t_D	t_M
42	0.50	1	1	2	3
43	0.50	1	1	2	2
42	0.75	1	1	2	3
43	0.75	1	1	2	2

Justification of the empirical boundary model. At the detection threshold, $\text{Power}(\alpha_c, T) = \pi^*$. Since α_c lies below the non-monotonic dip region ($\alpha_c < 0.55$), the signal model applies directly. From the power function $\text{Power} = \Phi(S(\alpha) \cdot \sqrt{T}/\sigma)$, setting $S(\alpha_c) \cdot \sqrt{T}/\sigma = z^*$ and substituting the signal model $S(\alpha) = c\,\alpha^\beta$ yields $c\,\alpha_c^\beta = z^*\sigma/\sqrt{T}$. Solving: $\alpha_c(T) = (z^*\sigma/c)^{1/\beta} \cdot T^{-1/(2\beta)}$, giving the stated form with exponent $\gamma = 1/(2\beta)$.

Calibration and consistency. Fitting $P(\text{det}|\alpha, T) = \Phi(A \cdot \alpha^\beta \cdot \sqrt{T} - z_0)$ to $T \in \{10, 20\}$ yields β=0.24, α_∞=0. Held-out test ($T \in \{5, 15\}$): MAE=0.106, 95% CI coverage 6/6. Leave-one-T-out CV (15 cells, sync-dip excluded): MAE=0.037, coverage 15/15. Profile likelihood yields $\hat{\beta}$=0.66 (CI $[0.24, 0.78]$); the wide interval reflects a flat surface.

Y Power Simulation: Shape Comparison

Simulated PermG detection under linear (H0), concave (H0′), and step (H1) signal models (1,000 reps, n_{perm}=1,000, BH-FDR q=0.05). Smooth models produce onset steepness k=4.9–6.1; the observed k=15.7 is ≈3× higher. The scaling exponent alone cannot distinguish smooth from step boundaries (H0′ γ=1.87 vs. observed γ=2.08). The 6/8 detection ceiling reflects MAUVE floor saturation, not signal limitations.

Z T=20 Extension Results

At $\alpha \geq 0.50$, PermG counts are consistent across T=10 and T=20 (6/8 at both), confirming signal-structure determination. At α=0.45, the shift from 0/8 (T=10) to 5/8 (T=20) supports power-limited (Model B) over structurally-undetectable (Model A), lowering α_c below 0.45. The α=0.60 dip persists at T=20 but dissolves at T=25 (5/8), confirming $\sqrt{T}$ power scaling eventually overcomes it. Directional selectivity is consistent: forward 6/8, reverse 0/8 at $\alpha \in \{0.50, 0.75, 1.0\}$ across all tested T.

Task degradation. End-of-training perplexity (gen 10, Pythia-410M, s42) improves at α=0.05 (20.8 vs. 26.6 baseline, mild augmentation effect) but degrades sharply between α=0.50 (31.3) and 0.52 (41.1), aligning with PermG detection onset.

Table 17: Cross-architecture PermG forward detection (α=1.0, 15 model-seed combinations across five families—Pythia, GPT-2, OPT, Qwen, SmolLM3—and two domains; Padding-410M is a Pythia variant). PermG fwd = count of FDR-significant forward pairs out of 8.

Model	Params	Domain	PermG fwd
Pythia-410M	410M	Wiki	**6/8**
Pythia-1.4B	1.4B	Wiki	2–6/8[†]
Pythia-6.9B	6.9B	Wiki	**7/8**[*]
GPT-2-medium	345M	Wiki	**6/8**[§]
OPT-1.3B	1.3B	Wiki	**6/8**
Qwen3.5-0.8B	0.8B	Wiki	**8/8**
SmolLM3-3B	3.0B	Wiki	3/8[‡]
Padding-410M	410M	Wiki	**6/8**
Pythia-1B	1.0B	C4	**8/8**
Pythia-1.4B	1.4B	C4	**8/8**

[†]Seed-dep. (s42: 2/8, s43: 6/8); 8-pair incl. MAUVE (floor ∼0.033 at 1.4B; 6-pair in Table 8). [*]bf16. [‡]ECE-only (ent. 0/4).
[§]s42 fp32: 6/8 fwd, 0 rev; s43: 6/8 both; bf16: 0/8.
2 seeds each for Pythia-410M/1.4B, GPT-2, OPT, Pythia-1B C4.

Z.1 Cross-Architecture Detection Table

Z.2 Cross-Architecture FPR Sensitivity

GPT-2-medium null-condition PermG FPR is 50% (6/12 pooled, 2 seeds), dramatically higher than Pythia-410M (7.4%), driven by ∼4× stronger null drift (entropy −20.5% vs. −4.9%; distinct-3 −35.8% vs. −4.9%). $K \geq 3$ consensus maintains 0% FPR (0/12) on GPT-2-medium, providing some protection, but standalone PermG requires architecture-specific null calibration.

Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.